



HINDUSTAN FORMULA KARTING CHAMPIONSHIP



SEASON-5

INNOVATION REPORT

<u>TEAM NAME :</u>	SPARKERS RACING	KART CATEGORY :	CV
TEAM VIN NO :	32	KART SPECIFICATION:	160CC
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ABSTRACT



This project introduces an innovative go-kart design combining mechanical improvements with a modern business model. a Japanese sheet under the chassis for enhanced strength and vibration reduction. It is equipped with ACB pedals (Accelerator, Clutch, Brake) to provide a professional driving feel. In addition, the business model is innovated through a Tap-Pay system for regular customers, along with a mobile app that enables ride scheduling and customer management. Together, these features enhance performance, safety, and user convenience, offering a practical solution for both racing and commercial karting applications.

INTRODUCTION

Go-karts are widely used for recreation and learning vehicle dynamics, but conventional designs often lack durability, safety, and customer-focused features. To overcome these gaps, our project introduces an innovative go-kart that combines mechanical improvements with a modern business model.

The kart uses straight and a Japanese sheet under the chassis for strength and vibration control.

ACB pedals (Accelerator, Clutch, Brake) give drivers a realistic automobile-like experience. On the business side, a Tap-Pay system enables quick payments for regular users, while a mobile app allows easy scheduling and ride management.

This approach makes the go-kart both technically advanced and commercially practical



PROBLEM STATEMENT

Traditional go-karts face challenges in safety, durability, and user convenience.

Basic pedal setups do not provide a realistic driving feel.

On the business side, most go-karts lack modern payment systems and customer scheduling solutions, making them less efficient for regular users and operators.

To overcome these issues, there is a need for a go-kart that combines mechanical innovations with digital features, improving performance, safety, and overall customer experience.



SOLUTION



A Japanese sheet under the chassis strengthens the structure, reduces vibrations, and increases durability.

ACB pedals (Accelerator, Clutch, Brake) are integrated to provide drivers with a realistic and professional driving experience.

A Tap-Pay system enables quick and seamless payments for regular customers.

A dedicated mobile application is developed for ride scheduling and customer management.

This combination of engineering design and smart technology ensures enhanced safety, performance, and customer convenience, making the go-kart both technically advanced and commercially viable.



JAPAN SHEET



Reduces vibrations from the track surface for smoother driving.

Protects the frame and components from stones, bumps, and debris.

Lightweight yet strong, avoids unnecessary weight gain.

Improves safety by acting as an underbody shield.

Extends the life of the chassis by reducing wear and damage.



ACB PEDALS



- Increased Driver Convenience: ACB pedal system provides comfort, control,
- Adjustable, customizable, and balanced pedal system prioritizes driver convenience and comfort
- Improved Safety: Enhanced stability and control reduce the risk of accidents.
- Ergonomic Design: ACB Pedal System reduces driver fatigue and improves comfort.



TAP PAY SYTEM & APPS

SCHEDULE



We introduced a Tap Pay system for regular customers in our go-karting facility to

Provide a faster, cashless, and hassle-free experience. Instead of waiting in long queues or handling physical payments, customers can simply use a preloaded smart card / RFID band / UPI-linked tap system to instantly pay for rides. This reduces waiting time, improves customer convenience, and allows regular riders to enjoy seamless access to the track. Additionally, it helps the management maintain digital records of rides, loyalty points, and discounts for frequent customers.

App-Based Scheduling – A mobile app where customers can pre-book their rides, select time slots, track availability, and receive notifications. This avoids waiting, ensures better crowd management, and gives customers a personalized racing experience.

Together, these innovations make go-karting faster, smarter, and more enjoyable for regular riders.

